DRILL 03

PPE & SCBA

NFPA 1001 Level 1 (JPR 4.3.1, 4.5.2) / WorkSafeNB Reg 91-191

SCBA CYLINDER REPLACEMENT (SOLO, BUDDY & CASCADE)

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Any Engine/Rescue | 🦺 **PPE:** Full Structural Gear or Work Uniform + Safety Glasses

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Firefighter Pairs

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to change an empty SCBA cylinder on their own pack in under 60 seconds, change their partner's cylinder while their partner remains on air with pack on (buddy swap), and safely operate the station cascade or compressor fill system. Rapid cylinder turnarounds keep working crews in service at working structure fires.

📦 TOOLS & APPARATUS SETUP

- 4 to 6 SCBA harnesses (seated in brackets or staged on floor).
- 6 to 8 spare SCBA cylinders (empty and full).
- Station cascade fill station / compressor (or mobile cascade if equipped).
- Safety glasses / eye protection and hearing protection for fill station.

⚠ SAFETY WATCH-OUTS & STOP RULE

⚠ **High-Pressure Hazard:** SCBA cylinders operate at 2,216 to 4,500 PSI. Never fill a cylinder without verifying hydrostatic test date and visual inspection for gouges, cracks, or burnt composite wrap.

⚠ **Blast Door Safety:** Never fill cylinders outside the certified containment fill station blast chamber. Blast door must be locked before opening fill valves.

🛑 **THE STOP RULE: Shout "STOP!" if someone attempts to fill a damaged or expired cylinder, or if high-pressure air leaks with a violent hiss.**

PHASE 1: INSPECTION & SOLO CYLINDER CHANGE (15 MINUTES) — SLOW DRY RUN

1 Inspection Points (Before Swapping):

- Inspect cylinder threads and the rubber high-pressure **O-ring** in the coupling. If O-ring is missing, nicked, or cracked, take cylinder out of service.
- Verify hydrostatic test date stamped on cylinder (Carbon fiber: 5-year cycle, 15-year total life).

2 Solo Swap Procedure (Pack on Ground):

- Close cylinder valve completely. Bleed residual air via regulator bypass.
- Disconnect high-pressure coupling (hand-tighten only, never use wrenches).
- Release cylinder retaining clamp/strap. Slide empty cylinder down and out.
- Insert full cylinder (valve facing down toward harness outlet).
- Lock retaining clamp. Hand-thread high-pressure coupling until snug.
- Open cylinder valve fully, listen for air leaks, check remote gauge vs cylinder gauge (must agree within 100–200 PSI).

DRILL 03: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

PPE & SCBA • Practical Application

PHASE 2: THE TWO-PERSON "IN-SERVICE" BUDDY SWAP (20 MINUTES) — BUDDY COACHING

1. Firefighter A kneels on one knee with full gear and SCBA on their back (staying on air).
2. Firefighter B stands behind:
   - Close Firefighter A's cylinder valve.
   - Firefighter A takes 2–3 breaths to bleed pressure until alarm chimes.
   - Firefighter B unscrews the high-pressure handwheel.
   - Firefighter B unbuckles retaining strap, lifts empty cylinder out.
   - Firefighter B inspects O-ring, slides full cylinder in, snaps retaining clamp shut.
   - Firefighter B connects high-pressure coupling hand-tight.
   - Firefighter B opens cylinder valve fully, taps Firefighter A on both shoulders, and announces: *"You're full! Valve open, 4,500 PSI, good to go!"*
3. Swap roles and repeat.

PHASE 3: STATION CASCADE FILL STATION OPERATION (15 MINUTES) — TIMED / STRESS

1. Review manufacturer SOP for station cascade:
   - Inspect cylinder for damage and test date.
   - Place cylinder inside fragmentation fill tube/chamber.
   - Connect fill whip and hand-tighten bleed screw.
   - Close and latch the heavy blast chamber door.
   - Select lowest bank cylinder that has higher pressure than the tank. Open slowly (max fill rate: 300–500 PSI/min to prevent heat buildup).
   - Cascade through Banks 1, 2, and 3 until target pressure (4,500 PSI) is reached.
   - Close bank valve, close cylinder valve, open bleed valve, disconnect whip.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Cross-threading high-pressure coupling:* Ruptures threads and destroys the connection. Always align straight and hand-thread smoothly.
- ❌ *Rapid filling:* Filling too quickly overheats the cylinder. When it cools down in the truck, pressure drops 500+ PSI, leaving you short on air!
- ❌ *Dropping cylinders on concrete:* Carbon-wrapped bottles can suffer invisible structural delamination. Never let cylinders tip over.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"How easy was it to guide the cylinder into the harness bracket while your partner was kneeling?", "Did we find any worn O-rings?"***

✅ APPARATUS RESET CHECKLIST

- ☐ All empty training cylinders refilled and placed back into apparatus racks.
- ☐ Cascade compressor turned off, bank valves isolated.
- ☐ Eye protection returned to storage.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS